

assignment-02-code

August 19, 2026

1 Assignment II

1.1 Import External Libraries

```
[1]: import numpy as np

from sklearn.preprocessing import normalize
```

1.2 2. Normalization Using Python

```
[2]: v1 = np.array([[2], [1]])
      print(v1)

      v2 = v1 / np.linalg.norm(v1)
      print(v2)
```

```
[[2]
 [1]]
[[0.89442719]
 [0.4472136 ]]
```

1.3 4. Normalization Using Python with Complex Coefficient

```
[3]: v1 = np.array([[2], [1+1j]])
      v2 = v1 / np.linalg.norm(v1)
      print(v2)
```

```
[[0.81649658+0.j          ]
 [0.40824829+0.40824829j]]
```

1.4 7. Vector Normalization by Hand and Using Python

```
[4]: # v1 = (1, 2) in the |0>/|1> basis
v1_01 = np.array([[1], [2]])

# basis change matrix from |0>/|1> to |+>/|-> coefficients
M = (1 / np.sqrt(2)) * np.array([[1, 1], [1, -1]])

v1_pm = M @ v1_01
print("v1 in +/- basis (unnormalized):", v1_pm.ravel())

v1_pm_normalized = v1_pm / np.linalg.norm(v1_pm)
print("v1 in +/- basis (normalized):", v1_pm_normalized.ravel())
```

```
v1 in +/- basis (unnormalized): [ 2.12132034 -0.70710678]
v1 in +/- basis (normalized): [ 0.9486833  -0.31622777]
```

1.5 9. Vector Normalization by Hand and Using Python 2

```
[5]: v1 = np.array([[1], [2]])
v1_normalized = v1 / np.linalg.norm(v1)
print("v1 normalized in 0/1 basis:", v1_normalized.ravel())

prob_0 = np.abs(v1_normalized[0, 0]) ** 2
print("P(|0>):", prob_0)
```

```
v1 normalized in 0/1 basis: [0.4472136  0.89442719]
P(|0>): 0.19999999999999998
```
